

Indian perspective

The Defence Research and Development Organisation's (DRDO) Daksh is a real proof of India's technological prowess. Daksh robot is India's first indigenously manufactured remotely-operated vehicle, which is capable of handling and disposing improvised explosive devices. After experimenting with around 20 units, the Indian army announced that it is procuring a hundred more units of Daksh for deploying in the army.

DRDO has also announced its plans to develop a robot soldier by 2020-30. "Whatever a soldier will do in warfare, a robot soldier should be able to do. That's the plan," said Dr V.K. Saraswat, scientific advisor to the defence minister and secretary of Defence R&D. The robot soldiers, controlled from remote locations, can do multiple tasks including fighting humans and carrying loads of ammunitions.

"We need to include a lot of artificial intelligence to avoid collision. Also, a lot of robot soldiers need to communicate with each other in the battlefield. Enormous amount of database and analytic intelligence is required for this," Dr Saraswat said. DRDO, apparently, also has plans to replace mules with robots to carry heavy loads to places like Siachen.

That said, there appears to be a bit of a disappointment in the local robotics ecosystem as most of the defence budget apart from the DRDO projects is going abroad. "There is huge potential growth in this space but the government needs to start investing in local companies like Israel, the US, Russia or China do, rather than depending on other countries for technology support, as it has traditionally been. As of now, it appears that 70 per cent of the defence budget of India is outsourced to other countries. There needs to be a change in the government's procuring process," comments Azad.

ics, especially their BigDog product. Imagine a huge dog with big strong legs carrying huge payloads and marching alongside soldiers as they cross mountainous or even desert regions! One great feature of the BigDog is force-controlled technology. With its quadruped gait, the robot dog, or mule as some call it, can regain balance if it is kicked, handle rough terrain like rocks, and climb inclines up to 35 degrees.

Last year, the company revealed a bigger and more useful beast code-named LS3 or BullDog. While the original BigDog could carry a payload of about 150 kg up to 20 km without having to refuel, the new model can carry 180 kg up to about 30 km. It is also quieter and can jump over obstacles, right itself after a fall and navigate with greater autonomy than its predecessor.

Growing intelligence

Despite the increasing physical capability and other benefits of robots, why are people still sceptical about their impending role in war? Is it simply because these are supposedly less intelligent than humans? While this is true to a large extent, the fact is that robots are slowly becoming more intelligent.

"The level of intelligence varies from robot to robot, depending on the mission

the robot is used for. A robot must at least be able to operate autonomously to the extent that the soldier can concentrate on the task at hand. It should act as a tool to reduce his load and not as a burden and technology barrier. In general, robots need not be highly intelligent but have to be smart, simple to use and maintain," opines Azad.

"Intelligence" in robotics is a very loosely-used term, but generally it revolves around the way a system responds to the feedback from the external environment and changes. Another challenge is the processing of data captured from various sensors used in the robot. In all cases, time in the order of milliseconds is of the highest priority, as lives depend on the response of these systems.

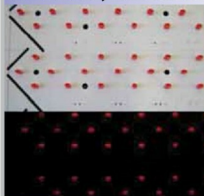
Nowadays, extremely complex software systems are being developed to improve the intelligence of robots. However, as far as military robots go, any such technology has to be tested time and again to prove its reliability before deployment because it is a matter of lives.

"Robots in defence, like the K-MAX, surely have extremely high levels of intelligence. Their learning algorithms are extremely effective, using various machine learning techniques. Most of these techniques are

Conformal coatings for LED application

ELPEGUARD®
PETERS

SL 1397, SL 1347



SL 1347 black-opaque

- for a strong contrast of the LEDs to the non-reflective substrate
- SL 1397 white-opaque
- for superior luminous efficacy
- excellent reflectivity ratings

Conformal coating spray



PETERS
Speziallacke für die Elektronik

Your local contact:
EMS Technologies
#G3 Reliance Residency
500029 Hyderabad.
Ph: 040-66567767

For Distribution Enquiries in all States contact: **+91-9133707031**
E-mail: info@emstechnologies.in
www.peters-india.com